

# Housing Finance Digest PDF Generation Flow

## Entry Point

Run app.py with --week-start, --week-end, --out and optional --export-json. app.py orchestrates article collection, scoring, LLM synthesis, newsletter composition and PDF rendering.

## Execution Order

- 1. config.py loads environment variables and competitor aliases. Required runtime settings include GEMINI\_API\_KEY for Gemini mode, optional Ollama settings, NEWS\_API\_KEY, and SQLITE\_DB\_PATH.
- 2. app.py creates ArticleStore from services/article\_store.py to persist fetched articles, prompts, LLM outputs, validation metrics and selected evidence into SQLite.
- 3. services/source\_feeds.py fetches RBI/NHB/CRISIL/ICRA policy and rating context articles from Google News RSS. These feed regulatory evidence and fallback regulatory rows.
- 4. services/news\_api.py uses NewsAPIClient to fetch competitor articles for each configured competitor and aliases. services/source\_feeds.py also adds supplemental RSS articles.
- 5. utils/url\_tools.py resolves Google News or aggregator URLs into canonical/original URL fields.
- 6. services/scoring.py annotates reliability, relevance, novelty and materiality; services/dedup.py clusters near-duplicate articles; services/scoring.py then finalizes scores and selects balanced evidence for LLM prompts.
- 7. services/prompts.py builds company intelligence prompts. services/gemini\_search.py runs the configured LLM provider and returns parsed JSON. models/schemas.py parses company-level JSON into WeeklyCompanySection objects.
- 8. app.py builds curated evidence for industry, regulatory and competitor sections. services/agentive\_pipeline.py normalizes and de-duplicates article outputs. services/section\_validators.py validates each section and may trigger repair prompts through the LLM.
- 9. services/prompts.py builds final section/final synthesis prompts; services/gemini\_search.py returns agentive analysis JSON. app.py post-processes weak regulatory rows using source-backed fallback logic where needed.
- 10. services/newsletter\_composer.py groups competitor rows, normalizes regulatory/industry structures, builds the newsletter\_composer prompt and calls the LLM once more to produce final newsletter JSON.
- 11. app.py adds default cover, index and section keys, writes optional \*\_agentive\_analysis.json and markdown summary, then passes newsletter\_raw to renderers/pdf\_renderer.py.
- 12. renderers/pdf\_renderer.py does not call any LLM. It consumes the final newsletter dictionary and uses ReportLab plus image\_bca198f4.png when available to write \*\_agentive\_analysis.pdf.

## Files Kept For Runtime

app.py, config.py, requirements.txt, image\_bca198f4.png, models/schemas.py, renderers/pdf\_renderer.py, services/article\_store.py, services/gemini\_search.py, services/news\_api.py, services/source\_feeds.py, services/scoring.py, services/dedup.py, services/prompts.py, services/agentive\_pipeline.py, services/newsletter\_composer.py, services/section\_validators.py, and utils/url\_tools.py.

## Files Removed As Non-runtime For This Flow

Old agents package, parsers package, docs, demos, README, .gitkeep, services/yfinance\_layer.py, and utils/company.py are not imported by app.py or pdf\_renderer.py in the traced PDF-generation path.